Features: Useful Clues

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected feature-choice record.

Choose inputs, break an insufficient record, challenge a shortcut and reconsider features for a changed task.

Keep one habit: Keep the target definition, available inputs and reasons for collecting them.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Categories and Labels

Pictures Become Numbers

Input, Process, Output

W1 · Define the usable feature record

> Separate the target and later sticker from pre-decision measurements.

Your evidence record

- For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality.

  Choose the measurements needed as **input**s from length, width, colour, card number and the final inspection sticker. State the **target label** and write the **feature record** for a 12 cm by 8 cm red card using a defined order.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Prove one measurement insufficient

> Construct equal-width cards with different labels.

Your evidence record

- Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Break a colour shortcut

> Test a deliberately different card against the stated dimensional rule.

Your evidence record

- A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels.

  Explain why the perfect practice score did not establish that colour was a reliable **feature** for this task.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Reconsider features when the task changes

> Use a contrasting pair to justify the newly needed property.

Your evidence record

- The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded **feature** becomes useful? Explain why length and width alone may now be insufficient.

  Give two cards that support your explanation.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Design a record for a new task

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Design a record for a new task

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Prove that one measurement is insufficient for **Feature**s and useful distinctions.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the **relevant feature** and explain why size is insufficient.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful **feature** and a distracting **feature**. Test your choice on a blue circle.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 8 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

A green acceptance sticker is attached only after the shelf check.

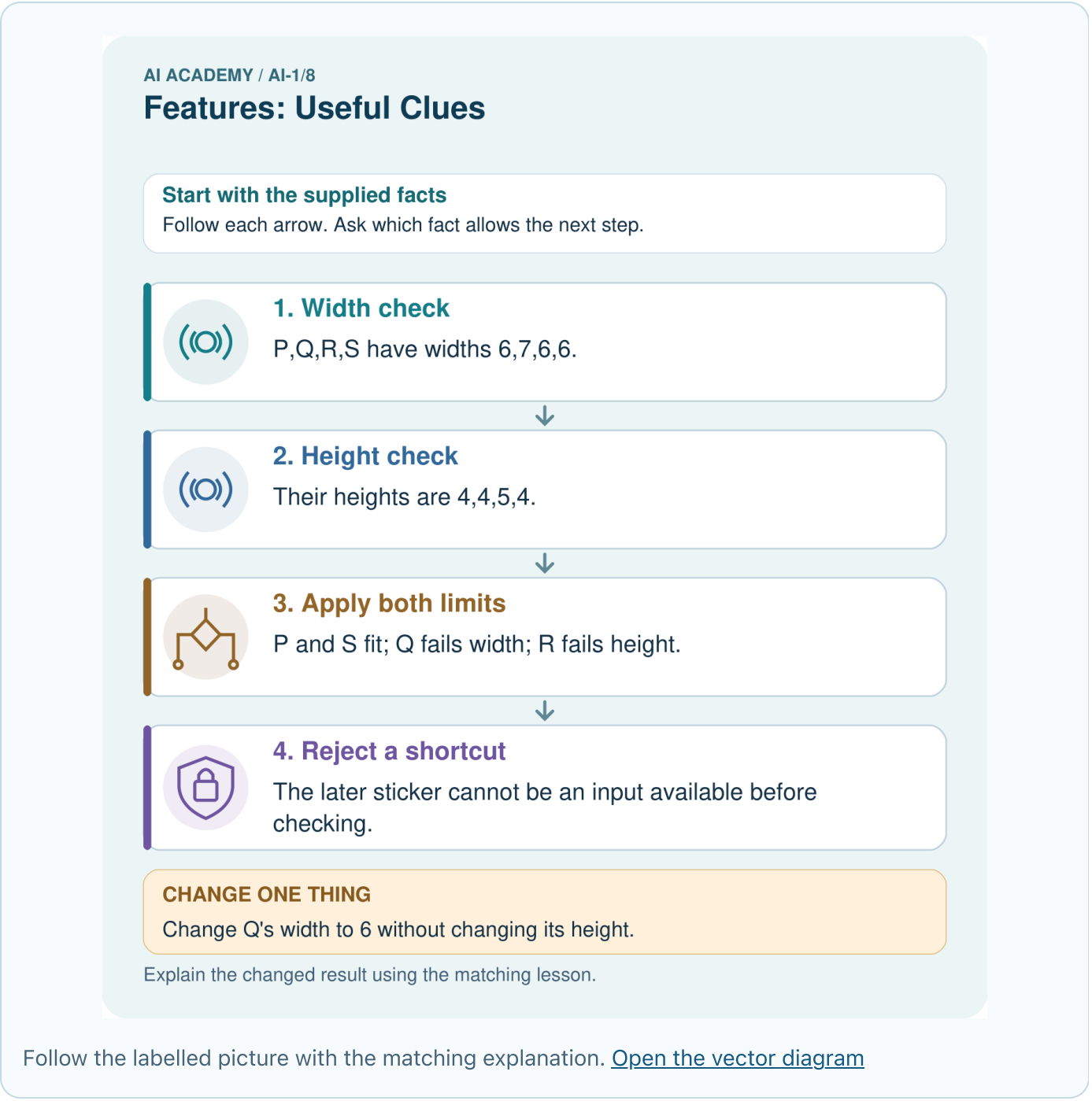
AI ACADEMY / AI-1/8

Dimensions determine shelf acceptance

	Width cm	Height cm	Outcome
P stripes	6	4	fits
Q stripes	7	4	width fails
R dots	6	5	height fails
S dots	6	4	fits

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.
- Which labelled step matches this detail? The later sticker cannot be an **input** available before checking.
- 1. Apply both limits
- 2. Reject a shortcut
- 3. Height check
- 4. Width check
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Reject a shortcut The later sticker cannot be an **input** available before checking. Comparing P with Q isolates width; comparing S with R isolates height.

Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**. A later acceptance sticker leaks the very answer the system should produce.

Changed-case question

- Changed-case practice: Change Q's width to 6 without changing its height. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Q fits the shelf.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Q fits the shelf. Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Repair and save your work

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Features: Useful Clues: the case

AI ACADEMY / AI-1/8 · 01 · LOOK AT THE FACTS

Features: Useful Clues: the case

The situation

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Supplied evidence

Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).
A green acceptance sticker is attached only after the shelf check.

By the end, I can choose task-relevant features and avoid a clue created after the decision.

Follow the labelled picture with the matching explanation. Open the vector diagram

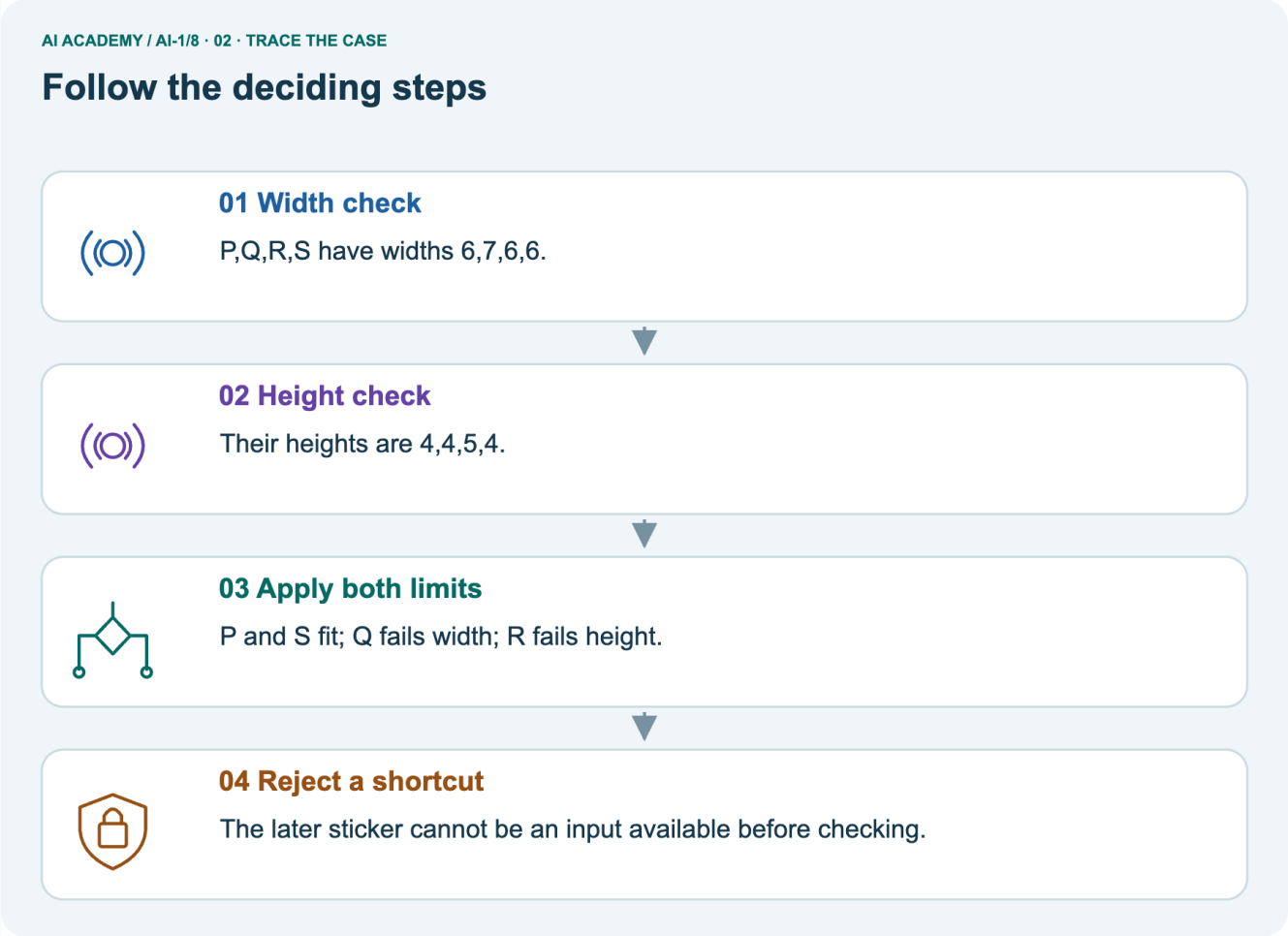
A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.
Supplied facts: Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

A green acceptance sticker is attached only after the shelf check.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



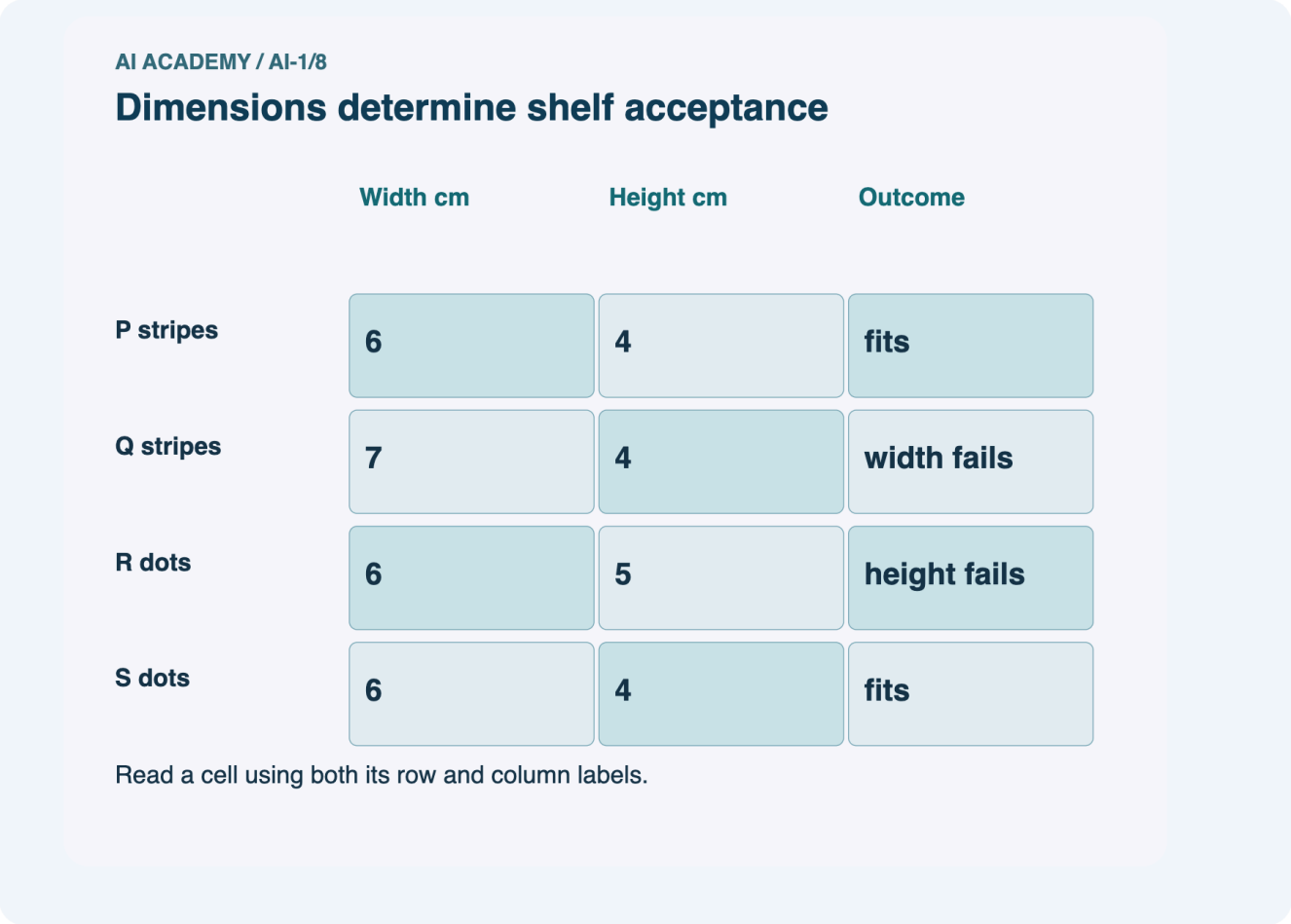
Follow the labelled picture with the matching explanation. Open the vector diagram

Width check: P,Q,R,S have widths 6,7,6,6. → Height check: Their heights are 4,4,5,4. → Apply both limits: P and S fit; Q fails width; R fails height. → Reject a shortcut: The later sticker cannot be an **input** available before checking.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Dimensions determine shelf acceptance



Follow the labelled picture with the matching explanation. Open the vector diagram

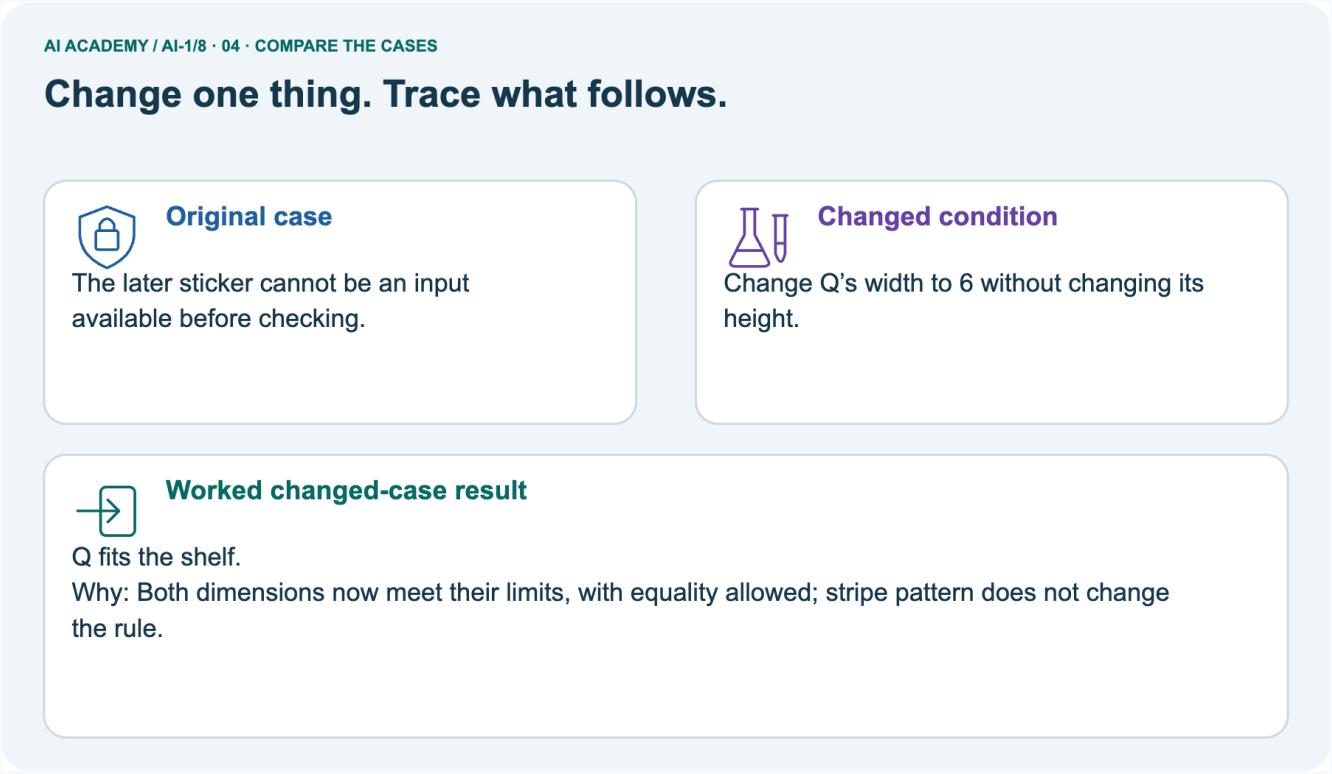
Dimensions determine shelf acceptance. P stripes: Width cm = 6, Height cm = 4, Outcome = fits. Q stripes: Width cm = 7, Height cm = 4, Outcome = width fails. R dots: Width cm = 6, Height cm = 5, Outcome = height fails.

S dots: Width cm = 6, Height cm = 4, Outcome = fits.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

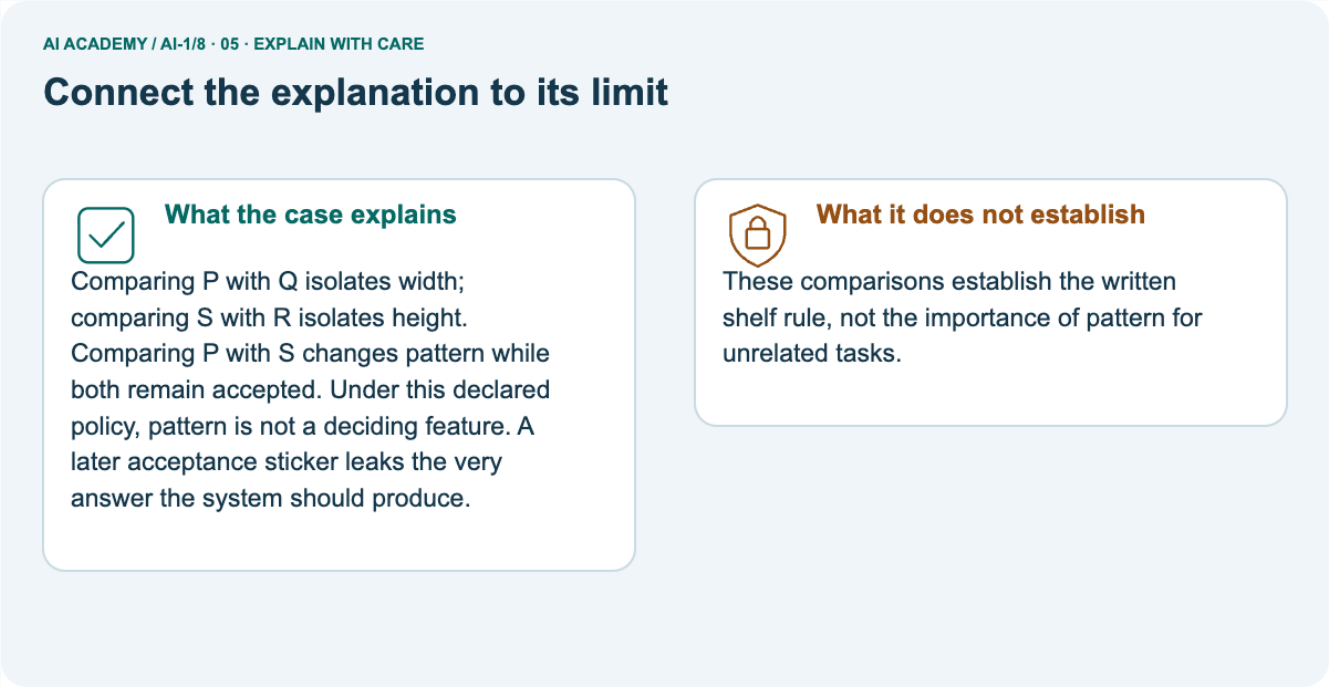
Original case: The later sticker cannot be an **input** available before checking. Changed condition: Change Q's width to 6 without changing its height. Expected result: Q fits the shelf.

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

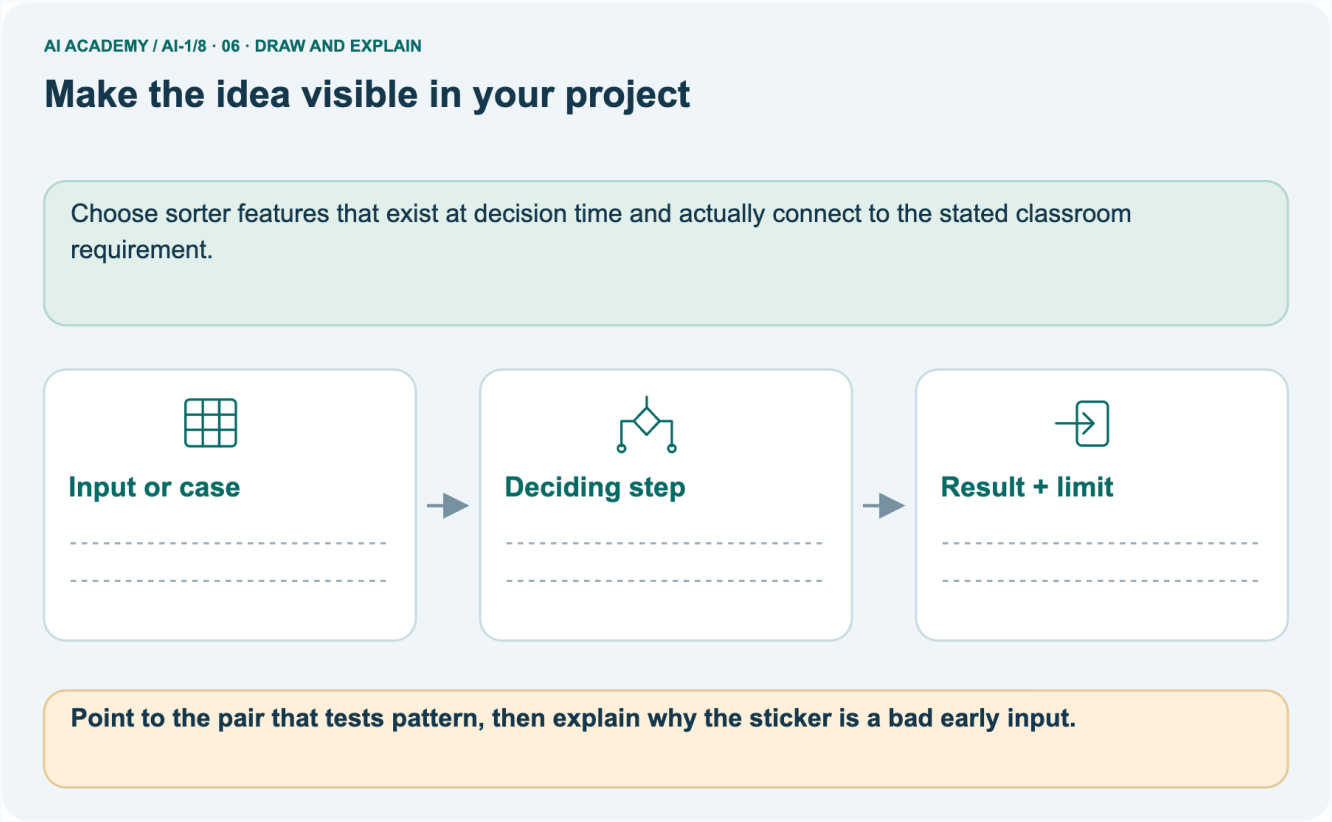
Comparing P with Q isolates width; comparing S with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce. Limit: These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement. Draw your **input**, deciding step, result and limit.

Supported question: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Features: Useful Clues: the case

02. Follow the deciding steps

03. Dimensions determine shelf acceptance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.